

CRISPR Gene Editing

CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) is a family of DNA sequences found in bacteria. CRISPR is part of an adaptive immune system that bacteria use to remember and destroy viruses they have encountered.

The CRISPR-Cas9 gene editing technique was developed in 2012 by Jennifer Doudna at the University of California, Berkeley, and Emmanuelle Charpentier at the Max Planck Institute in Berlin. Their paper, published in Science in August 2012, showed that Cas9 could be programmed with a guide RNA to cut DNA at any desired location.

Feng Zhang at the Broad Institute and George Church at Harvard independently demonstrated CRISPR-Cas9 editing in human cells in January 2013. A patent dispute between Berkeley and the Broad Institute lasted years, with the US Patent Trial and Appeal Board ruling in 2022 that the Broad's claims were distinct.

Doudna and Charpentier were awarded the 2020 Nobel Prize in Chemistry. The first approved CRISPR therapy, Casgevy for sickle cell disease, was approved by the UK Medicines and Healthcare products Regulatory Agency in November 2023.